

Newman-Penrose Formalism Calculations

Stephani et al., Equation (13.76)

Metric

$$ds^2 = -dv \otimes dv - 2kydv \otimes d\varphi + \frac{1}{t^2k^2}dt \otimes dt + \frac{1}{t^2k^2}dy \otimes dy - 2kyd\varphi \otimes dv + (-4k^2y^2 + t^2)d\varphi \otimes d\varphi$$

Coordinates

$$(x^\mu) = (v, t, y, \varphi)$$

Null Tetrad

Covariant components

$$l = \left(-\frac{1}{2}\sqrt{2}\right) dv + \left(\frac{\sqrt{2}}{2t^{(k^2)}}\right) dt + \left(-\sqrt{2}ky\right) d\varphi$$

$$n = \left(-\frac{1}{2}\sqrt{2}\right) dv + \left(-\frac{\sqrt{2}}{2t^{(k^2)}}\right) dt + \left(-\sqrt{2}ky\right) d\varphi$$

$$m = \left(\frac{\sqrt{2}}{2t^{(k^2)}}\right) dy + \left(\frac{1}{2}i\sqrt{2}t\right) d\varphi$$

$$\bar{m} = \left(\frac{\sqrt{2}}{2t^{(k^2)}}\right) dy + \left(-\frac{1}{2}i\sqrt{2}t\right) d\varphi$$

Contravariant components

$$l = \left(\frac{1}{2}\sqrt{2}\right) dv + \left(\frac{1}{2}\sqrt{2}t^{(k^2)}\right) dt$$

$$n = \left(\frac{1}{2} \sqrt{2} \right) dv + \left(-\frac{1}{2} \sqrt{2} t^{(k^2)} \right) dt$$

$$m = \left(-\frac{i \sqrt{2} k y}{t} \right) dv + \left(\frac{1}{2} \sqrt{2} t^{(k^2)} \right) dy + \left(\frac{i \sqrt{2}}{2 t} \right) d\varphi$$

$$\bar{m} = \left(\frac{i \sqrt{2} k y}{t} \right) dv + \left(\frac{1}{2} \sqrt{2} t^{(k^2)} \right) dy + \left(-\frac{i \sqrt{2}}{2 t} \right) d\varphi$$

Directional Derivatives

$$D = \frac{1}{2} \sqrt{2} \partial_v + \frac{1}{2} \sqrt{2} t^{(k^2)} \partial_t$$

$$\Delta = \frac{1}{2} \sqrt{2} \partial_v - \frac{1}{2} \sqrt{2} t^{(k^2)} \partial_t$$

$$\delta = -\frac{i \sqrt{2} k y}{t} \partial_v + \frac{1}{2} \sqrt{2} t^{(k^2)} \partial_y + \frac{i \sqrt{2}}{2 t} \partial_\varphi$$

$$\bar{\delta} = \frac{i \sqrt{2} k y}{t} \partial_v + \frac{1}{2} \sqrt{2} t^{(k^2)} \partial_y - \frac{i \sqrt{2}}{2 t} \partial_\varphi$$

Spin Coefficients

$$\rho = \frac{\sqrt{2} k^2 t^{(k^2)}}{4 t} + \frac{i \sqrt{2} k t^{(k^2)}}{2 t} - \frac{\sqrt{2} t^{(k^2)}}{4 t}$$

$$\sigma = \frac{\sqrt{2} k^2 t^{(k^2)}}{4 t} + \frac{\sqrt{2} t^{(k^2)}}{4 t}$$

$$\kappa = 0$$

$$\mu = \frac{\sqrt{2}k^2t^{(k^2)}}{4t} + \frac{i\sqrt{2}kt^{(k^2)}}{2t} - \frac{\sqrt{2}t^{(k^2)}}{4t}$$

$$\lambda = \frac{\sqrt{2}k^2t^{(k^2)}}{4t} + \frac{\sqrt{2}t^{(k^2)}}{4t}$$

$$\nu = 0$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = \frac{i\sqrt{2}kt^{(k^2)}}{4t}$$

$$\gamma = \frac{i\sqrt{2}kt^{(k^2)}}{4t}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{k^2 t^2 k^2}{t^2}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = 0$$

$$\Lambda = 0$$

Weyl Tensor Components

$$\Psi_0 = -\frac{i k^3 t^2 k^2}{2 t^2} - \frac{i k t^2 k^2}{2 t^2}$$

$$\Psi_1 = 0$$

$$\Psi_2 = \frac{i k^3 t^2 k^2}{2 t^2} - \frac{k^2 t^2 k^2}{t^2} - \frac{i k t^2 k^2}{2 t^2}$$

$$\Psi_3 = 0$$

$$\Psi_4 = -\frac{i k^3 t^2 k^2}{2 t^2} - \frac{i k t^2 k^2}{2 t^2}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "D"